

Module Code: PDAN8411

Title: Part 1 – Linear Regression

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Table of Contents

Introduction	3
Analysis planning	3
Exploratory Data Analysis (EDA)	4
Data Visualization	4
Feature Selection	8
Model Development	9
Model training.....	9
Model Evaluation	10
Conclusion.....	10

Introduction

In the evolving landscape of healthcare, understanding the diverse needs of clients is crucial for medical aid schemes. Recently, a prominent medical aid scheme in South Africa reached out with a vision to enhance their service offerings. They aim to tailor their medical aid costs based on the unique lifestyles and geographic regions of their clients. To achieve this, I have been tasked with developing a linear regression model that will provide a sliding scale of charges. This model will take into account several key factors, including age, sex, BMI, number of children, smoking habits and geographical region. By quantifying the relationship between these factors and healthcare expenses, the model will enable the scheme to implement a more equitable and data-informed pricing structure. I will be conveying my data analysis plan, actions and findings through this report.

Analysis planning

The below are the steps that I will be following as a plan on how I will be conducting my analysis:



Exploratory Data Analysis (EDA)

Data overview:

The insurance US-based dataset has been used for the purposes of analysing the data, cleaning data and developing a linear regression model. The information, features or variables used are contained in a file named "insurance_csv", it contains individual information regarding their medical costs. Below I have a table to framework the data that is contained in the dataset including columns, description and column data types.

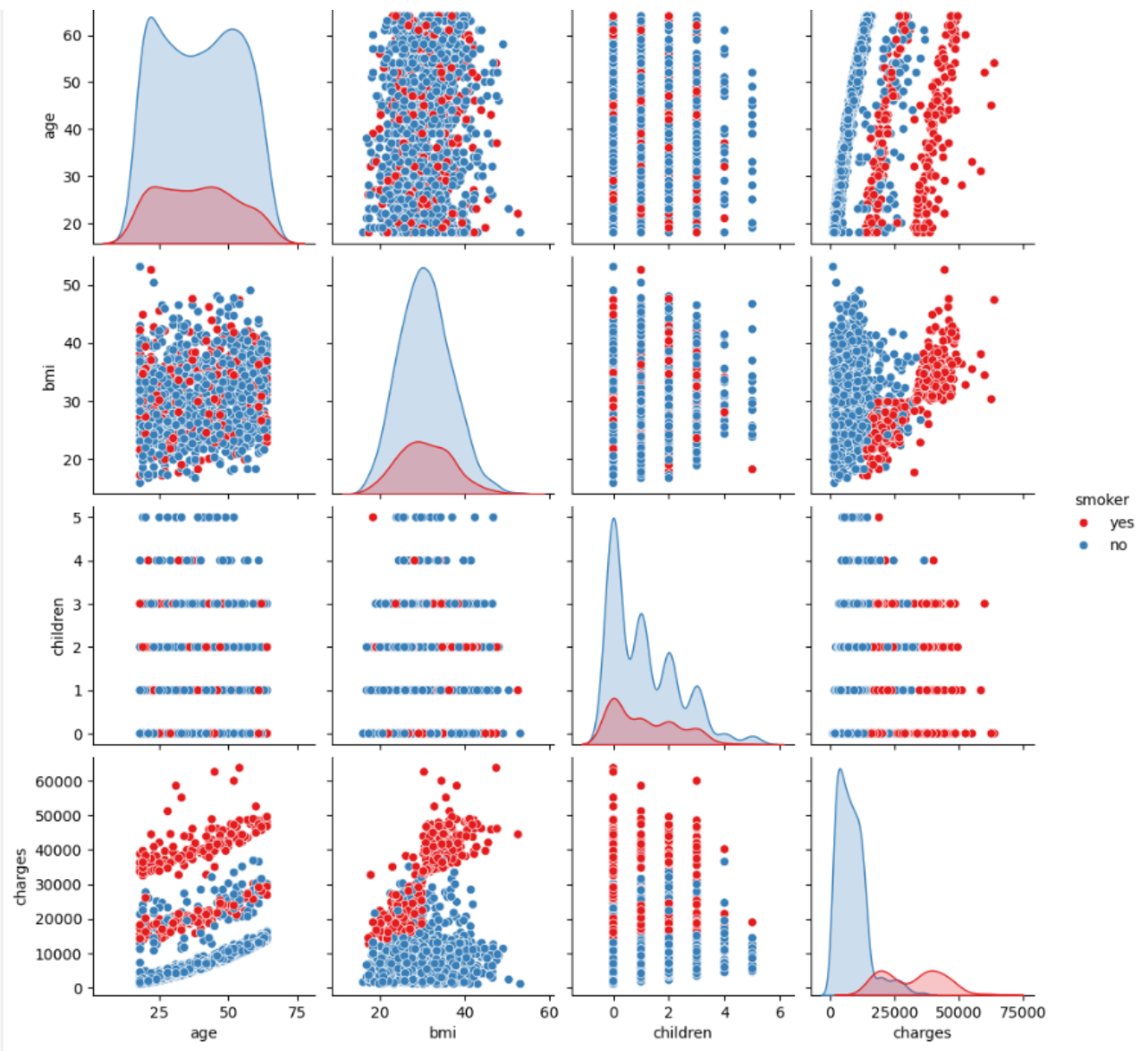
Column	Description	Data type
age	Age of primary beneficiary	int64
sex	Gender (male or female) of primary beneficiary	object
bmi	Body mass index of the primary beneficiary	float64
children	Number of dependents covered by health insurance	int64
smoker	Indicates whether primary beneficiary is a smoker or not (yes or no)	object
region	Primary beneficiary's residential location/area (northeast, southeast, southwest or northwest)	object
charges	Individual medical costs billed by health insurance	float64

The dataset has 7 columns in total, it has 6 independent variables and 1 dependent variable being "charges". There are 1338 entries (Rows & Columns). There are 3 numerical types being age, bmi and charges. There are also 3 categorical types being sex, smoker and region. Dataset does not have any missing values.

Before proceeded to data visualisation, I had examined the dataset to check for total number of rows contained and if there were any duplicates or missing values. After performing the examination of the dataset, I had found out that it has 1338 entries, there were no missing values and there was 1 duplicate that I thereafter dropped. I also examined the different categories of categorical variables / columns separately.

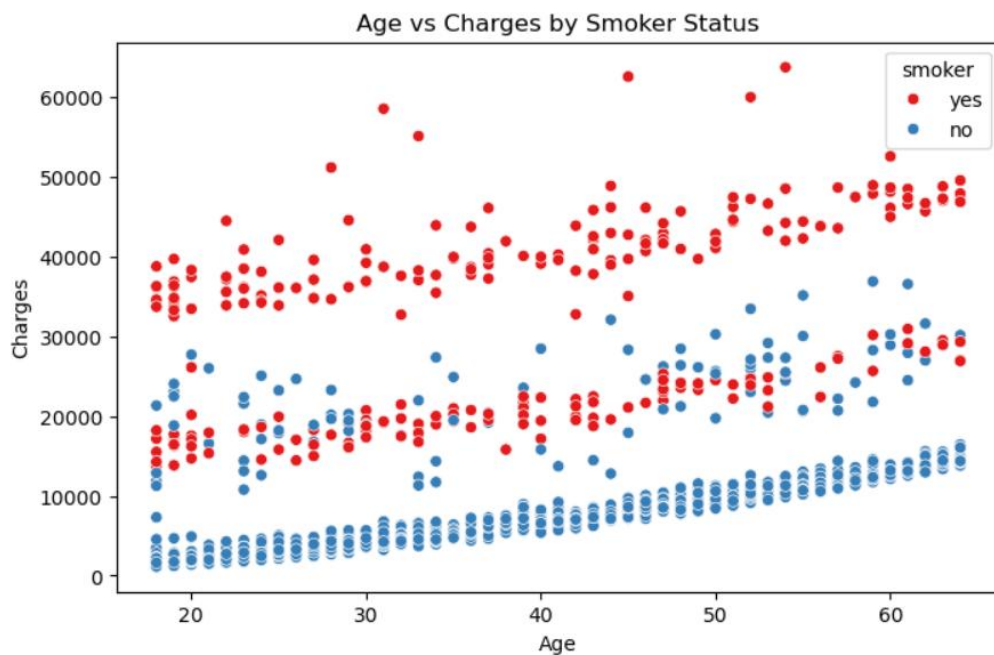
Data Visualization

To ensure that the information is fully understood, I had done some data visualization which is the graphical representation of information and data using visual elements such as charts and graphs to help understand patterns, trends, and insights.

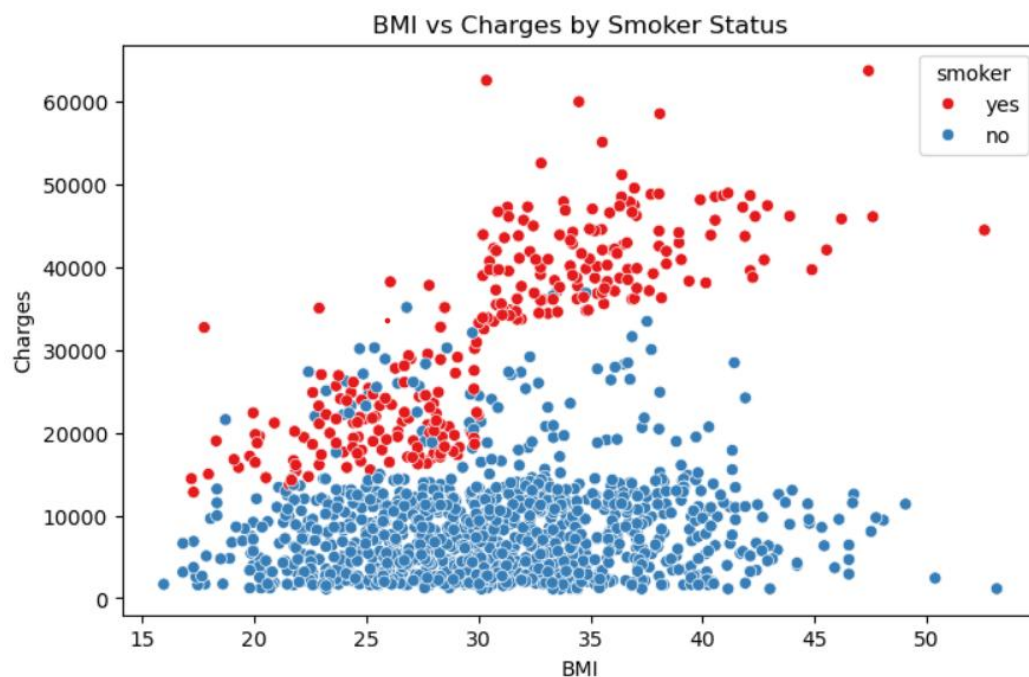


Above is a pairplot, this visualization helps in understanding the relationships between multiple variables simultaneously, it helps in visualizing how multiple variables relate to each other. Smokers are concentrated in the higher charges range. Smoking is the strongest cost driver. Higher BMI often corresponds to higher charges, especially for smokers. Non-smokers with high BMI still cost more than low BMI clients. Clients with more children show slightly higher charges. The trend is weaker compared to age or BMI. Number of dependents has a small positive effect on costs. Dataset is suitable to apply a linear regression model.

The scatterplot below shows a positive trend – as age increases, charges also increase. Age is a strong predictor and smoker status amplifies costs. People who smoke have higher medical charges than people who do not smoke.

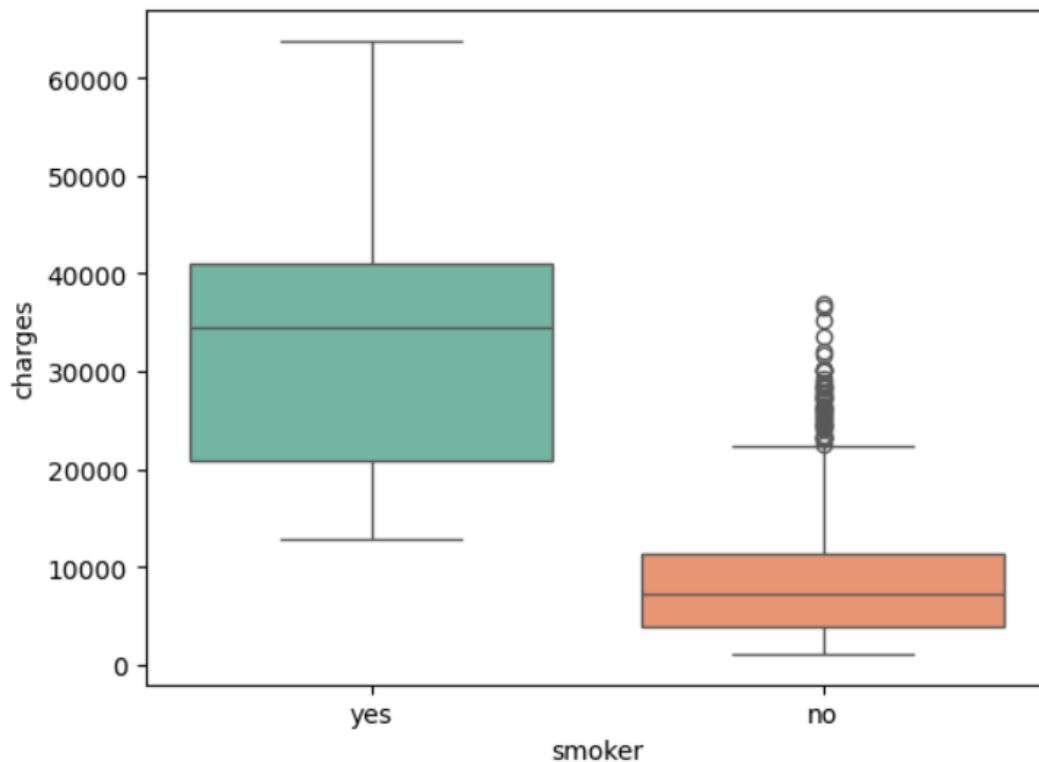


On the below scatterplot, there is a slight upward trend – higher BMI often corresponds to higher charges. BMI is a moderate predictor and its effect interacts with smoker status.

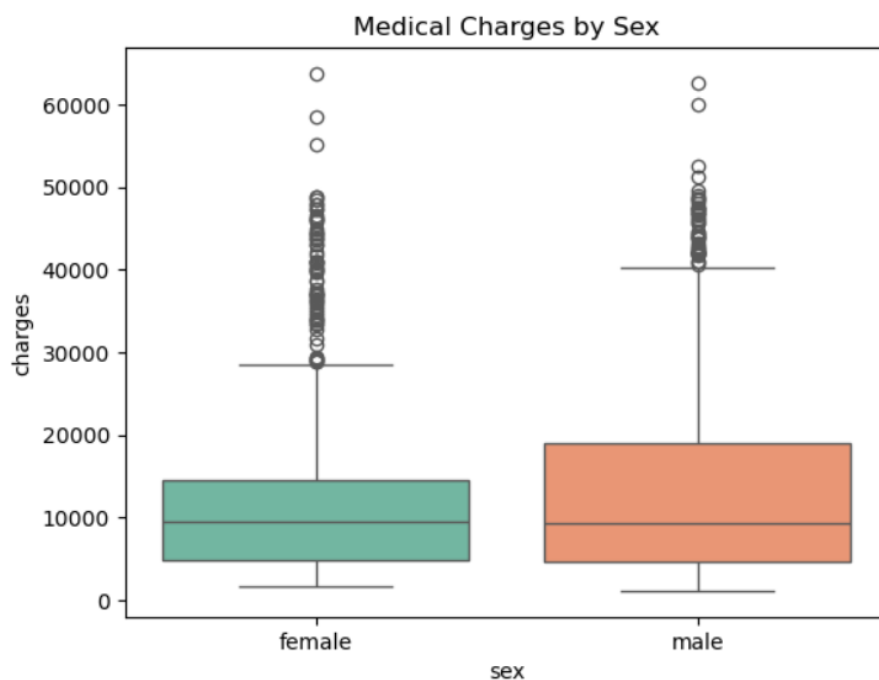


Next I will do the boxplot visualizations:

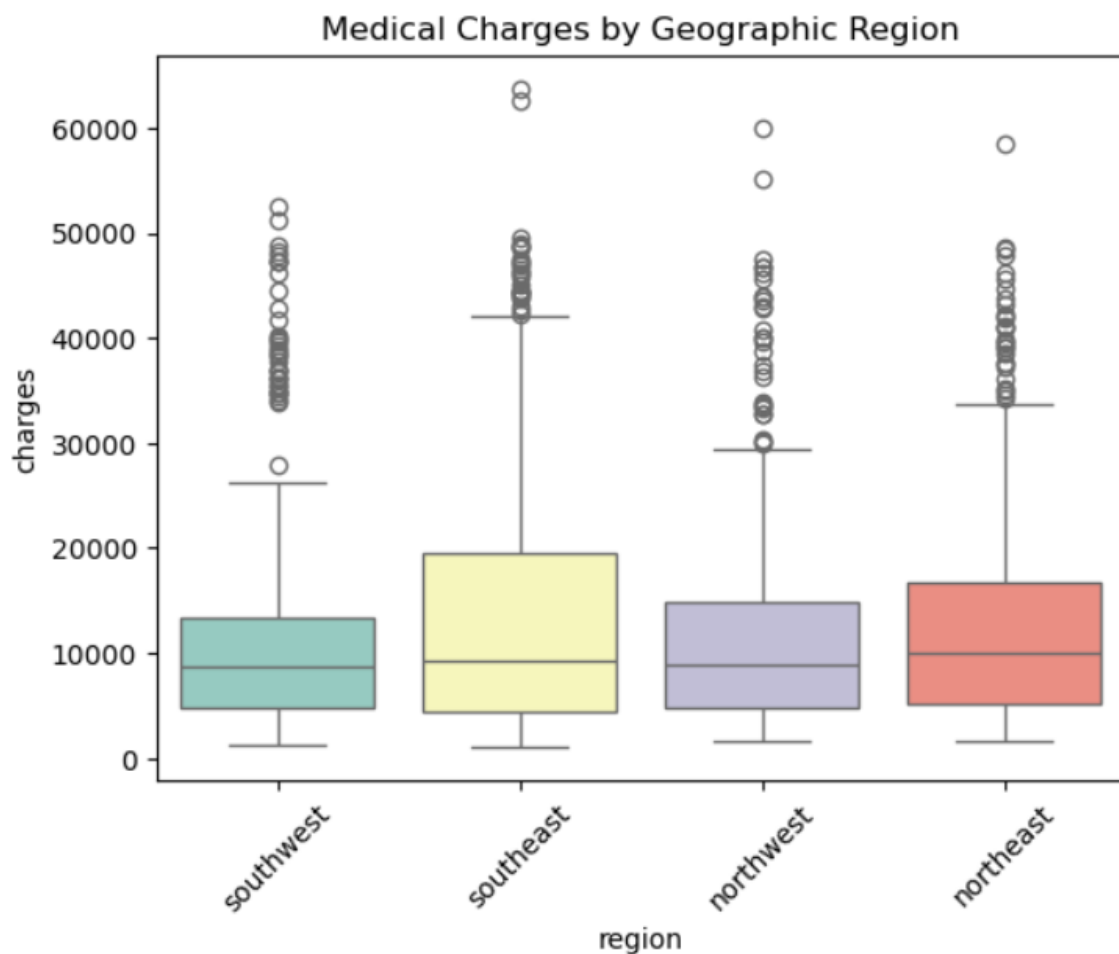
This plot shows Smoker vs Charges, smokers have a much higher median charge than non-smokers. Some smokers have extremely high charges, forming a long upper whisker. Smoking is the strongest cost driver, it significantly increases medical aid charges.



This plot shows Sex vs Charges, there is minimal difference between males and females in median charges. Sex has a small effect on charges.

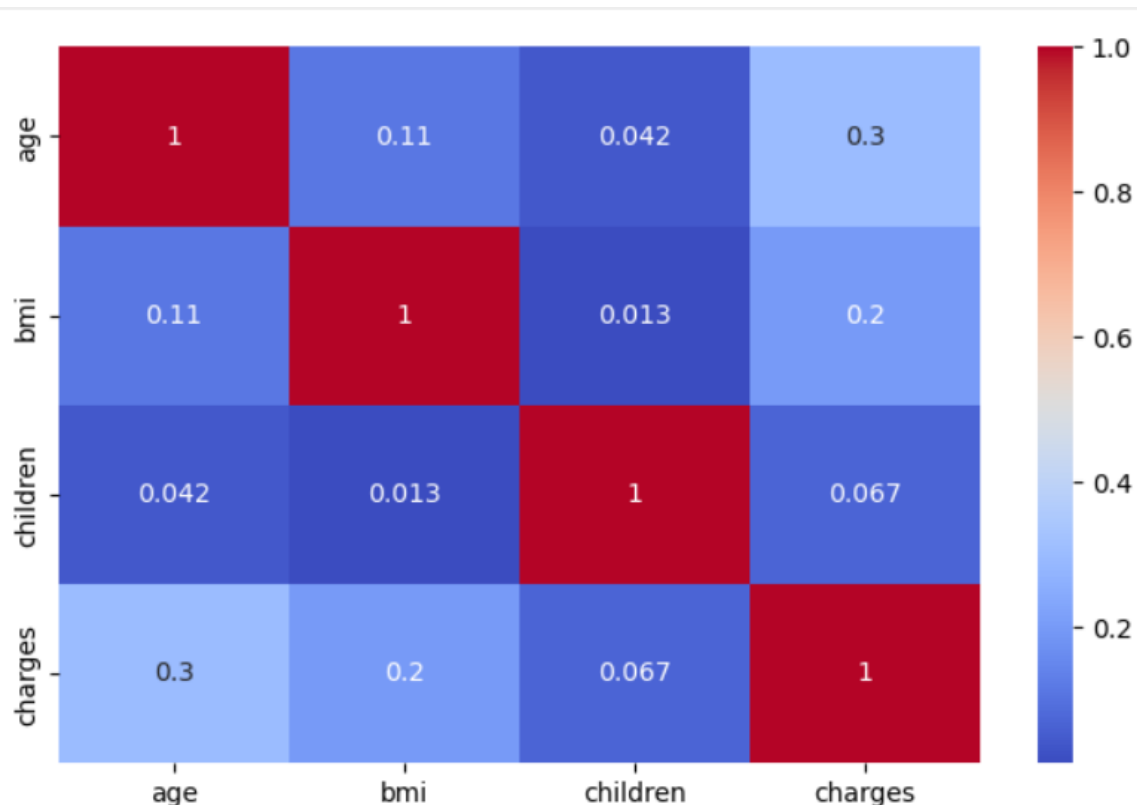


This plot shows Region vs Charges, All regions have high-cost outliers but the median differences indicate regional variation in costs. Region is a moderate predictor; it reflects geographic differences in healthcare costs.



Feature Selection

Feature selection helps identify which variables have the most significant impact on insurance charges and for this I used correlation analysis. Since mathematical equations cannot 'read' text like 'Smoker' or 'Region', I will convert these categorical variables into a numerical format and this is done by means of encoding.



Smokers pay significantly higher insurance costs, older individuals tend to have higher charges while BMI and Number of dependents have only a small effect.

Model Development

In reference to the EDA above, it is clear that there is a linear relationship between the categorical variable “age” and the numerical variable “charges” which is also a dependent variable. I also discovered that as the “age” increases, the “charges” are also increasing. Therefore the dataset is suitable for a linear regression model. Linear regression is used to model the relationship between a dependent target and one or more independent predictors/features.

Model training

To train the model I used Linear Regression. This allowed me to see how several independent variables like age, BMI, and smoking status work together to predict the dependent variable which is a medical charges.

Model Evaluation

My linear regression performance is shown by the values below:

```
Linear regression performance:  
Root Mean Squared Error (RMSE): 5956.34  
R-squared (R2 Score): 0.8069
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80.69% of the variance in insurance charges is explained by the predictors. This is good, indicating the model captures most of the variability.

Conclusion

The linear regression model was trained to predict insurance charges based on predictors such as age, BMI, smoker status, number of children, sex, and region. The model explains 80.69% of the variance in charges ($R^2 = 0.8069$), indicating a good fit for the data. Smoker status is the strongest predictor, followed by age and BMI. The linear regression model is solid for identifying main factors affecting insurance charges.